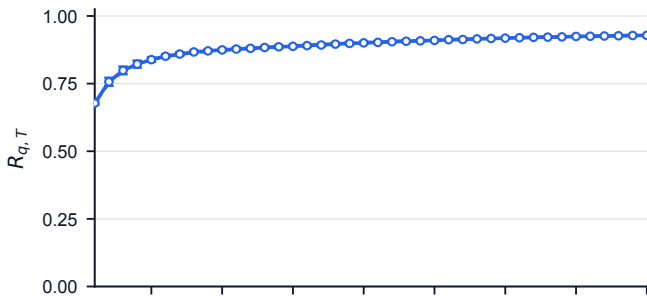
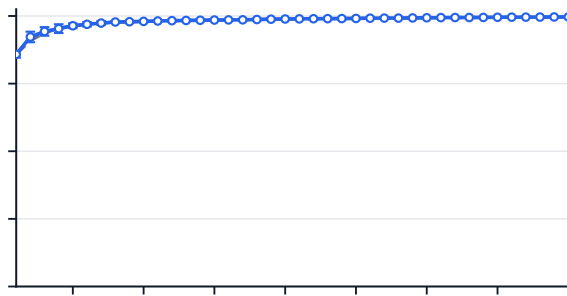


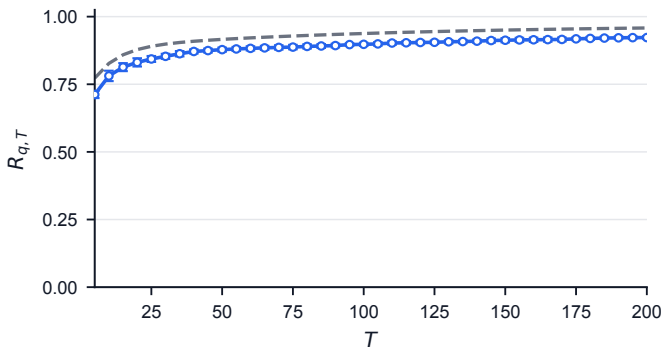
(a) 1-hop hyperedges



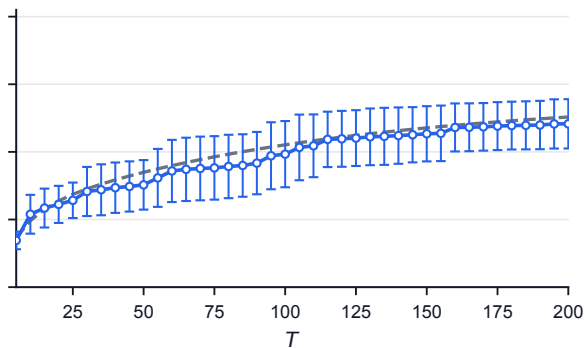
(b) 2-simplices



(c) Cycle-basis 2-cells



(d) Cycle-span 2-cells



$q = 8, n = 10$



Empirical mean $\pm$ 1 s.d.



Expected